

INTRODUCTION

Electronic Piano circuit using 555 timer IC and a few other electronics components.

This circuit consists of a number of push button switches (similar to different keys in a piano) which, when pressed, produce sounds with different tones via a speaker.

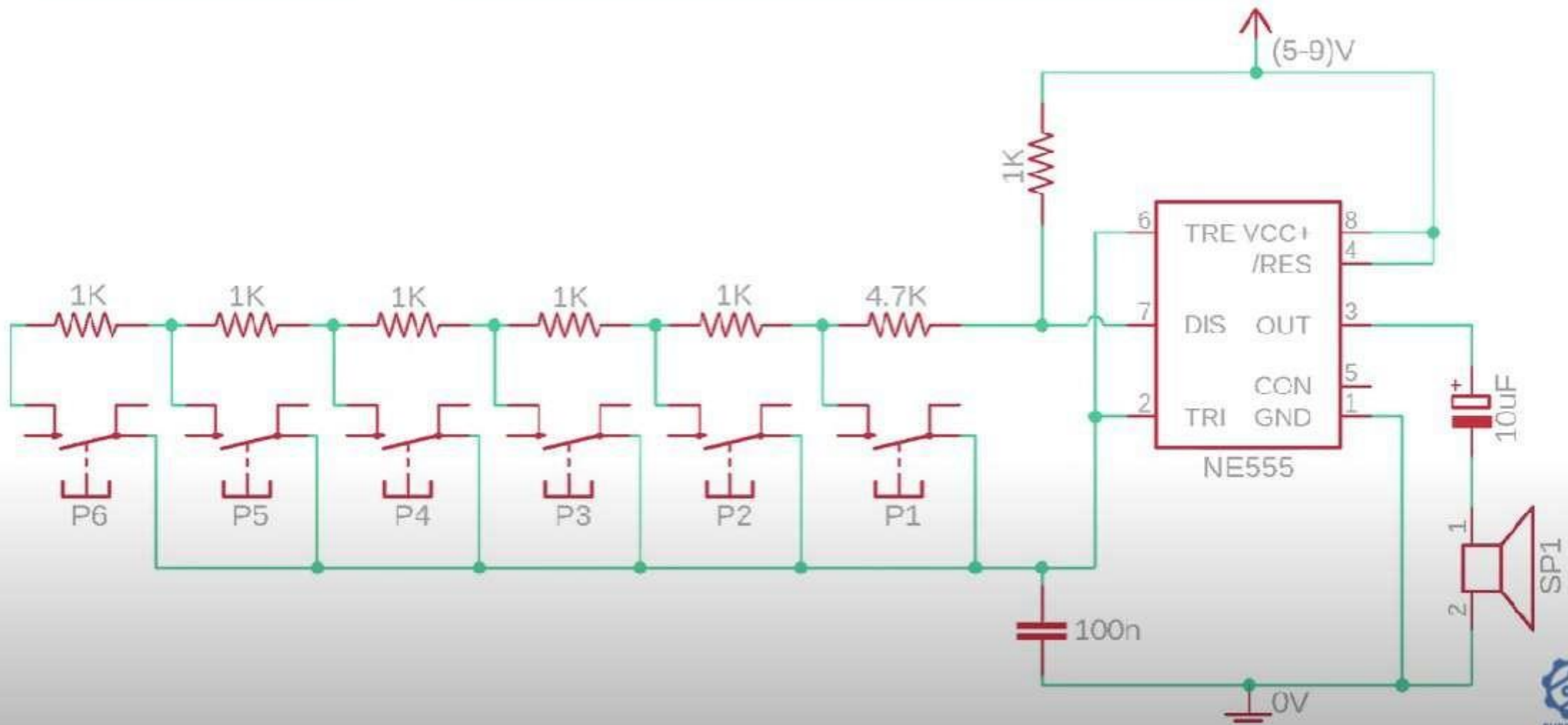
The frequency of the output tones can also be fine tuned by calculating the precise values of resistors to use.

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Working Principal

In this circuit, the 555 timer IC will work as an Astable multivibrator producing a square wave of different frequencies at the output. The speaker will produce different sounds depending upon the frequency of the signal. To change the frequency of the output signal, vary the total resistance of the circuit by pressing different switches. So every time you press a button a different amount of resistance gets connected to generate a different frequency square wave and the speaker will produce different tones.

Pin 6 and Pin 2 will allow retriggering of 555 Timer IC after every timing cycle. So we will connect these pins together to the ground through a capacitor. The output pin 3 is connected to the speaker through a 10uF capacitor and the fourth pin is connected to the VCC to avoid any sudden resets.



Sr. No.	Name of component	Specification	Quantity	Cost per piece (in Rs)
1	Resistance	1k	6	3
		4.7 k	3	3
2	IC	555 IC	3	8
3	Push button	Momentary push button	12	6
4	Capacitors	100uf	3	10
		10uF	3	8
5	Speaker	8 ohm	2	20
6	Power supply	9 Volt	1	100
7	Breadboard	A breadboard is used to make up temporary circuits for testing or to try out an idea.	1s	70

COMPONENTS DETAILS

1. IC -The 555 timer IC is an integral part of electronics projects. Be it a simple project involving a single 8-bit micro-controller and some peripherals or a complex one involving system on chips (SoCs), 555 timer working is involved. These provide time delays, as an oscillator and as a flip-flop element among other applications. Depending on the manufacturer, the standard 555 timer package includes 25 transistors, 2 diodes and 15 resistors on a silicon chip installed in an 8-pin mini dual-in-line package (DIP-8)

2. CAPACITORS:

1. 100 μF -Electrolytic decoupling capacitors 100 μF /25V. These capacitors are great transient/surge suppressors. Attach one between the power and ground of your project to ensure smooth power delivery. High quality radial electrolytic capacitors.
2. 10 μF -since the cap (short in the electronic world for capacitors) is rated for 10 μF , it can **hold a charge of ten micro coulombs** (that is, ten millionths of a Coulomb, 0.000010 C) per volt of voltage across its terminals

3.RESISTORS-: (1) 1k Ω resistor has a value of **1,000 ohms** and the number we will code is 1,000.1K-Commonly used in **breadboards and other prototyping applications**, these 1K ohm resistors make excellent pull-ups, pull-downs and current limiters. These thick-lead versions of the resistors fit snugly into a breadboard with very little movement

(2)4.6K-A passive device that resists the flow of electricity. This resistor will provide 4.7K Ohms of resistance wherever it is placed and will handle 1/4 watts. This resistor serve as a great pull up resistor and for open collector and open drain lines.

WORKING PRINCIPLE

How This Circuit Works

we have learnt how changing the value of resistor in series with the capacitor changes the flashing rate (Frequency). And the relation is of inverse proportionality if the value of resistance in series to the capacitor is large, the frequency will be less, and vice versa. You may refer to that tutorial for better understanding of how the values of resistors and capacitor influence the output frequency.

Formula for calculating the frequency of the output sound/tone:

$$\text{Frequency (f)} = 1.44 / ((R1 + 2 * R2) \times C1)$$

In the circuit we made, the value of R1 is 1K and C1 is 100n. After replacing these values in the above formula, we have:

$$\text{Frequency (f)} = 1.44 / ((1000 + 2 * R2) \times 10^{-7})$$

We can also rearrange the formula to calculate the value of resistor we need to use for any frequency required:

$$\text{Resistance (R2)} = ((1.44 \times 10^7) / f) - 500$$

Result:-

Astable multivibrator using 555 timer IC was built and fabricated on PCB. The frequency of o/p waveform is and duty cycle is .

Bibliography:-

Astable Multivibrators, IC-555 timer.

Digital Design by M. Morris Mano and Michael D. Ciletti

IC-555 Piano Idea

<https://www.instructables.com/circuits/howto/ic+555+piano/>

Astable multivibrator using 555 IC

<https://www.electronicshub.org/astable-multivibrator-using-555-timer/>